The gamma-product limit for the classical Bohnenblust–Hille estimates

A full solution of the conjecture in arXiv:1205.4735

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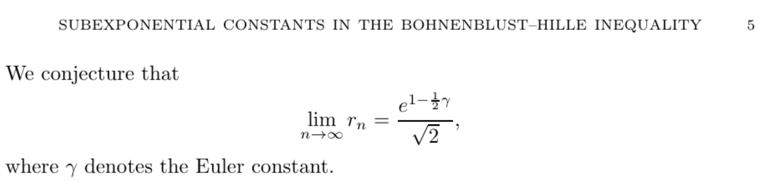
Status. Candidate full solution, likely valid, pending human review. The exact corrected gamma product converges to the conjectured constant, with an explicit first-order logarithmic error. The convergence premise left open in arXiv:1107.4814 is therefore also proved.

1 Source conjecture and provenance

Muñoz-Fernández, Pellegrino, and Seoane-Sepúlveda introduced explicit Bohnenblust–Hille upper estimates involving an even-indexed gamma product r_n, but left its convergence unproved [1]. Serrano-Rodríguez subsequently corrected a finite-index imprecision and, on PDF page 5, conjectured that the corrected product satisfies [2]

$$\lim_{n\to\infty} r_n = \frac{\mathrm{e}^{1-\gamma/2}}{\sqrt{2}},$$

where n tends to infinity through the even integers and γ is the Euler–Mascheroni constant.


SUBEXPONENTIAL CONSTANTS IN THE BOHNENBLUST–HILLE INEQUALITY 5

We conjecture that

$$\lim_{n\to\infty} r_n = \frac{e^{1-\frac{1}{2}\gamma}}{\sqrt{2}},$$

where γ denotes the Euler constant.

Figure 1: The complete conjecture on source PDF page 5 of arXiv:1205.4735.

For even $n > 14$, the corrected formula (equation (2.3) of the source) is

$$r_n = \frac{\pi^{(n^2-196)/(8n)}}{2^{(n^2-2n-192)/(4n)}\left[\prod_{k=7}^{(n-2)/2} \Gamma\left(\frac{6k+1}{4k+2}\right)^{2k+1}\right]^{1/n}}. \tag{1}$$

2 Proof intuition

The apparently complicated product is a Cesàro average in disguise. The gamma arguments obey

$$\frac{6k+1}{4k+2} = \frac{3}{2} - \frac{1}{2k+1}.$$

After inserting $\Gamma(3/2) = \sqrt{\pi}/2$, all terms of macroscopic size cancel exactly. Each remaining summand is

$$(2k+1)\left[\log\Gamma\left(\frac{3}{2} - \frac{1}{2k+1}\right) - \log\Gamma\left(\frac{3}{2}\right)\right],$$

which tends to $-\psi(3/2)$. Its average therefore gives the limit, and the next Taylor term gives the sharper $-(\psi_1(3/2)/4)(\log n)/n$ correction. The 2012 correction deletes only six product factors and changes powers by $O(1/n)$ in the logarithm, so it cannot affect the limit or its logarithmic leading error.

3 Full theorem

Write $\psi = \Gamma'/\Gamma$ and $\psi_1 = \psi'$. Let r_n be the corrected quantity in (1).

Theorem 1. *As $n \to \infty$ through the even integers,*

$$\log r_n = 1 - \frac{\gamma}{2} - \frac{1}{2}\log 2 - \frac{\psi_1(3/2)}{4n}\log n + O\left(\frac{1}{n}\right). \tag{2}$$

Consequently

$$\lim_{n\to\infty} r_n = \frac{\mathrm{e}^{1-\gamma/2}}{\sqrt{2}}.$$

The same asymptotic expansion holds for the original, uncorrected product in arXiv:1107.4814.

Proof. It is convenient first to use the original product

$$\widetilde{r}_n = \frac{\pi^{(n^2-4)/(8n)}}{2^{(n-2)/4}\left[\prod_{k=1}^{(n-2)/2}\Gamma\left(\frac{6k+1}{4k+2}\right)^{2k+1}\right]^{1/n}}. \tag{3}$$

Put $m = (n-2)/2$ and

$$f(x) = \log\Gamma\left(\frac{3}{2} - x\right), \qquad f_0 = f(0).$$

Since $\Gamma(3/2) = \sqrt{\pi}/2$ and

$$\sum_{k=1}^{m}(2k+1) = m(m+2) = \frac{n^2-4}{4},$$

the powers of π and the constant gamma contributions cancel exactly in the logarithm of (3). What remains is the identity

$$\log\widetilde{r}_n = \frac{n-2}{2n}\log 2 - \frac{1}{n}\sum_{k=1}^{m}(2k+1)\left[f\left(\frac{1}{2k+1}\right) - f_0\right]. \tag{4}$$

Taylor's theorem at zero gives, uniformly as $t \to \infty$,

$$t\,[f(1/t) - f_0] = f'(0) + \frac{f''(0)}{2t} + O(t^{-2}). \tag{5}$$

Here

$$f'(0) = -\psi(3/2), \qquad f''(0) = \psi_1(3/2).$$

Moreover,

$$\sum_{k=1}^{m} \frac{1}{2k+1} = \frac{1}{2}\log n + O(1), \qquad \sum_{k=1}^{m} \frac{1}{(2k+1)^2} = O(1).$$

Substitution of (5) in (4), using $m/n = 1/2 - 1/n$, yields

$$\log \widetilde{r}_n = \frac{1}{2}\log 2 + \frac{1}{2}\psi(3/2) - \frac{\psi_1(3/2)}{4n}\log n + O(n^{-1}). \tag{6}$$

The classical identity

$$\psi(3/2) = 2 - \gamma - 2\log 2$$

turns the constant in (6) into $1 - \gamma/2 - (\log 2)/2$.

It remains to check the corrected formula rather than merely the original one. Direct comparison of (1) and (3) gives the exact identity

$$\log r_n - \log \widetilde{r}_n = \frac{\kappa}{n}, \tag{7}$$

where the constant independent of n is

$$\kappa = 48\log 2 - 24\log\pi + \sum_{k=1}^{6}(2k+1)\log\Gamma\left(\frac{3}{2} - \frac{1}{2k+1}\right).$$

Indeed, the corrected powers differ by $-24/n$ for π and $-48/n$ in the denominator exponent of 2, while its product omits precisely the indices $1,\ldots,6$. Equation (7) transfers (6) unchanged at its displayed precision and proves the theorem. $\square$

4 Upgrade: the formerly conditional ratio limit

The source arXiv:1107.4814 also considered the real recursive estimates

$$C_2 = \sqrt{2}, \quad C_3 = 2^{5/6}, \qquad C_n = \sqrt{2}\left(\frac{C_{n-2}}{A^2_{(2n-4)/(n-1)}}\right)^{(n-2)/n} \qquad (n > 3), \tag{8}$$

where A_p is the sharp lower Khinchin constant. The paper derived a one-step ratio limit conditional on convergence of its r_n. The same local calculation proves a stronger unconditional asymptotic for the entire recursive sequence.

Corollary 1. *For the estimates in* (8)*,*

$$\log C_n = \frac{n}{8}\log 2 + 1 - \frac{\gamma}{2} - \frac{1}{4}\log 2 - \frac{\psi_1(3/2)}{4n}\log n + O\left(\frac{1}{n}\right). \tag{9}$$

In particular,

$$\frac{C_n}{2^{n/8}} \longrightarrow \frac{\mathrm{e}^{1-\gamma/2}}{2^{1/4}}, \qquad \frac{C_n}{C_{n-1}} \longrightarrow 2^{1/8}.$$

These statements concern the explicit upper estimates C_n, not the unknown optimal Bohnenblust–Hille constants.

Proof. Set $s_n = n \log C_n$. Once $p_n = (2n-4)/(n-1)$ lies in Haagerup's gamma branch,

$$A_{p_n} = \sqrt{2} \left(\frac{\Gamma(3/2 - 1/(n-1))}{\sqrt{\pi}} \right)^{1/p_n}.$$

All earlier indices contribute only a bounded parity-dependent constant to s_n. Multiplying the logarithm of (8) by n and using $f_0 = \frac{1}{2}\log \pi - \log 2$ gives, for all sufficiently large n,

$$s_n - s_{n-2} = \left(\frac{n}{2} + 1\right) \log 2 - (n-1) \left[f\left(\frac{1}{n-1}\right) - f_0 \right]. \tag{10}$$

Sum (10) separately on the even and odd subsequences. On either parity,

$$\begin{aligned} \sum_{j \le n} \left(\frac{j}{2} + 1\right) &= \frac{n^2}{8} + \frac{3n}{4} + O(1), \\ \#\{j \le n\} &= \frac{n}{2} + O(1), \\ \sum_{j \le n} \frac{1}{j-1} &= \frac{1}{2} \log n + O(1), \end{aligned}$$

where all sums are restricted to the relevant parity and may begin at any fixed sufficiently large index. Applying (5) in (10) therefore gives

$$s_n = \frac{n^2}{8} \log 2 + n \left(\frac{3}{4} \log 2 - \frac{1}{2} f'(0) \right) - \frac{f''(0)}{4} \log n + O(1).$$

Divide by n, insert $f'(0) = -\psi(3/2)$ and $f''(0) = \psi_1(3/2)$, and use the displayed digamma identity. This is (9). Its error tends to zero, so subtracting the formulas for n and $n-1$ proves the ratio limit. □

5 Verification, scope, and novelty

The proof uses only exact algebra, Taylor's theorem for $\log \Gamma$ near $3/2$, the elementary harmonic-sum asymptotic, and the standard identities $\Gamma(3/2) = \sqrt{\pi}/2$ and $\psi(3/2) = 2 - \gamma - 2\log 2$. The included verifier checks the finite-correction identity to high precision, the predicted first-order error, and the recursive corollary. These computations are sanity checks, not proof steps.

The result completely answers the explicit conjecture in arXiv:1205.4735 and removes the convergence hypothesis attached to the classical recursive estimates in arXiv:1107.4814. It does *not* determine the optimal Bohnenblust–Hille constants; later work already replaced these classical estimates by much slower-growing families.

A bounded novelty audit on 11 August 2026 searched the run's four lightweight indexes; exact web phrases containing the conjectured constant and r_n; Crossref; OpenAlex's eight works then recorded as citing DOI `10.1016/j.laa.2012.11.028`; and locally available arXiv full text for the citing Bohnenblust–Hille corpus (including arXiv:1205.2385, 1207.0124, 1302.3026, 1503.07306, 1506.00159, and 1604.06323). No proof of the exact gamma-product conjecture was found. ArXiv:1108.1550 computes the related single-factor Khinchin limit that explains the same constant, but does not state or prove convergence of this product. Novelty is plausible, not certified.

Human-review recommendation. Check the exact cancellation in (4), the factor 1/4 in the trigamma correction, and the finite correction (7). For the upgrade, independently check the increment identity (10). If these pass, classify the conjecture itself as fully solved and the recursive asymptotic as a valid strengthening.

References

[1] G. A. Muñoz-Fernández, D. Pellegrino, and J. B. Seoane-Sepúlveda, *Estimates for the asymptotic behavior of the constants in the Bohnenblust–Hille inequality*, Linear Multilinear Algebra **60** (2012), 573–582; arXiv:1107.4814, doi:10.1080/03081087.2011.613833.

[2] D. M. Serrano-Rodríguez, *Improving the closed formula for subpolynomial constants in the multilinear Bohnenblust–Hille inequalities*, Linear Algebra Appl. **438** (2013), 3124–3138; arXiv:1205.4735, doi:10.1016/j.laa.2012.11.028.